Assignment 1: Code Comparison

Basic Info:

I used Copilot and Claude to generate my code. I accessed Copilot from Google and Claude on my computer. I used the prompt: "Objective Write a C program that picks a secret number between 1 and 10. The user gets up to 3 tries to guess the number. Requirements • Use a fixed secret number (for example, 7). • Use printf to prompt the user and scanf to read their guess. • Use a while or for loop to let the user guess up to 3 times. • Use if-else statements to check if the guess is too high, too low, or correct. • Stop the loop immediately if the user guesses the correct number. • Print a final win or lose message. Example Output text Guess a number between 1 and 10. Attempt 1/3. Enter your guess: 5 Too low! Try again. Attempt 2/3. Enter your guess: 8 Too high! Try again. Attempt 3/3. Enter your guess: 7 Correct! You win!"

Code they generated:

Claude:

```
#include <stdio.h>


int main() {
    int secret = 7;
    int guess;
    int attempts = 3;
    int won = 0;

    printf("Guess a number between 1 and 10.\n");

    for (int i = 1; i <= attempts; i++) {
        printf("Attempt %d/%d. Enter your guess: ", i, attempts);
        scanf("%d", &guess);

        if (guess == secret) {
            printf("Correct! You win!\n");
```

```
            won = 1;
            break;
        } else if (guess < secret) {
            printf("Too low! Try again.\n");
        } else {
            printf("Too high! Try again.\n");
        }
    }

    if (!won) {
        printf("You lose! The correct number was %d.\n", secret);
    }

    return 0;
}
```

Copilot:

```
#include <stdio.h>

int main() {
    int secret = 7;     // fixed secret number
    int guess;
    int attempt;
```

```
    printf("Guess a number between 1 and 10.\n");

    for (attempt = 1; attempt <= 3; attempt++) {
        printf("Attempt %d/3. Enter your guess: ", attempt);
        scanf("%d", &guess);

        if (guess == secret) {
            printf("Correct! You win!\n");
            return 0;   // stop immediately
        } else if (guess < secret) {
            printf("Too low! Try again.\n");
        } else {
            printf("Too high! Try again.\n");
        }
    }

    // If loop finishes without a correct guess:
    printf("Sorry, you lose. The correct number was %d.\n", secret);

    return 0;
}
```

e. Select one to use as the basis for this assignment and justify why you picked it. Explain what you did to improve it in terms of correctness, execution time, space complexity, and maintainability.

Analysis:

In terms of correctness, neither handles cases where the user enters a string. Both just use up each guess then ends the program. They both have similar execution times as they use the same for loops and if else statements. However, Clause adds an extra if at the end. If the user doesn't win it prints a statement while Copilot losing statement runs if the program isn't ended by the user winning. When it comes to space complexity Copilot is slightly more efficient. Claude uses 4 base variables where Copilot uses 3. Claude then makes another variable "i" in its for loop where Copilot uses its previously assigned variable "attempts." In maintainability they both had very few comments explaining the code so it may be hard for future users to make adjustments.

I will use the code created by Copilot. It has very few adjustments to make and would be the easiest to edit. To improve the correctness, I would allow it to handle cases where the user inputs a string. Since Copilot doesn't have the extra if statement, there isn't anything I could improve. It already uses the most efficient methods. Likewise, it also is already more efficient when it comes to space complexity as it uses less variables and uses all of them efficiently. Lastly, for maintainability I would add more comments to explain how the code works.